

Scalability & Future Plan

Date	06 July 2026
Team ID	
Project Name	Voice Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, ensuring long-term viability and impact.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Whisper transcription and SBERT scoring run synchronously in a single FastAPI process, handling one request at a time.	High latency and blocked requests when multiple users submit recordings at the same time.	High
2	Sessions (audio + PDF reports) are stored as local files under temp/sessions/ with no automatic cleanup or database.	Unbounded disk growth over time and no persistence if the server restarts or is redeployed.	Medium
3	Only 10 predefined concepts are available; users cannot add their own topics or reference material.	Limits the tool's use to a fixed set of tech topics, reducing applicability across other subjects.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single FastAPI process handles requests one at a time; no queueing.	Introduce an async task queue (e.g. Celery/Redis) with multiple worker instances behind a load balancer to process analyses in parallel.
2	Data Storage	Audio files and PDF reports saved to local disk under temp/sessions/.	Move to cloud object storage (S3/GCS) for audio and reports, with a database (PostgreSQL) for session metadata and a TTL-based cleanup job.
3	Performance	Whisper 'base' model and SBERT inference run on CPU, taking several seconds per request.	Use GPU-backed inference, cache loaded models across requests, and evaluate faster Whisper variants (e.g. faster-whisper) to cut latency.
4	Security	CORS is open to all origins (*) and API endpoints have no authentication or rate limiting.	Restrict CORS to known origins, add API key/JWT authentication, apply rate limiting, and encrypt stored audio and reports.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-language support and custom concept/reference upload.	Q4 2026	Broadens usage beyond English and the 10 fixed tech topics.
Phase 3	Real-time streaming analysis with live feedback during recording.	Q1 2027	Instant feedback improves the practice experience and engagement.
Phase 4	Mobile app and LMS integration (e.g. Moodle/Canvas) for institutional use.	Q3 2027	Enables adoption at classroom/institution scale.